

INTRODUCTION TO OBJECT ORIENTED PROGRAMMING

Methods and Properties

1) Create an Employee class with the following specifications:

- 1. Employee Name as string.**
- 2. Basic Salary, HRA, DA, TAX, Gross Pay and Net Salary as decimal.**
- 3. Calculate the HRA (15% of Basic Salary), DA (10% of Basic Salary), Gross Pay (Basic Salary + HRA + DA), Tax (8% of Gross Pay) and Net Pay (Gross Pay – Tax).**
- 4. A Constructor to allow to define the Employee Name and Basic Salary.**
- 5. A method CalculateNetPay() to calculate the HRA, DA, Tax, Gross Pay and Net Pay values.**
- 6. A method Display() to display the salary structure.**

```
using System;
```

```
class Employee
```

```
{
```

```
    string empName;
```

```
    decimal basicSalary, hra, da, tax, grossPay, netPay;
```

```
    public Employee(string name, decimal salary)
```

```
    {
```

```
        empName = name;
```

```
        basicSalary = salary;
```

```
    }
```

```
    public void CalculateNetPay()
```

```
    {
```

```
        hra = basicSalary * 0.15m;
```

```
        da = basicSalary * 0.10m;
```

```
        grossPay = basicSalary + hra + da;
```

```
        tax = grossPay * 0.08m;
```

```
        netPay = grossPay - tax;
```

```
    }
```

```
    public void Display()
```

```
    {
```

```
        Console.WriteLine("\n----- Employee Salary Details -----");
```

```
        Console.WriteLine("Employee Name : " + empName);  
        Console.WriteLine("Basic Salary : " + basicSalary);  
        Console.WriteLine("HRA      : " + hra);  
        Console.WriteLine("DA      : " + da);  
        Console.WriteLine("Gross Pay  : " + grossPay);  
        Console.WriteLine("Tax      : " + tax);  
        Console.WriteLine("Net Pay   : " + netPay);  
    }  
}
```

```
class Program
```

```
{  
    static void Main()  
    {  
        Console.Write("Enter Employee Name: ");  
        string name = Console.ReadLine();  
  
        Console.Write("Enter Basic Salary: ");  
        decimal salary = Convert.ToDecimal(Console.ReadLine());  
  
        Employee emp = new Employee(name, salary);  
  
        emp.CalculateNetPay();  
    }  
}
```

```
emp.Display();  
  
Console.ReadLine();  
}  
}
```

OUTPUT:

Enter Employee Name: Ravi

Enter Basic Salary: 50000

----- Employee Salary Details -----

Employee Name : Ram

Basic Salary : 50000

HRA : 7500.00

DA : 5000.00

Gross Pay : 62500.00

Tax : 5000.00

Net Pay : 57500.00

2) Create a new class called "Stock" which contains the following:

- 1. A String for the stock name.**
- 2. A String for the stock symbol.**

3. A double storing the previous closing price of the stock.

4. A double storing the current closing price of the stock.

5. A Constructor that allows you to define a stock's name, symbol, previous closing price and current closing price.

6. A method `GetChangePercentage()` that returns the percentage change from the previous closing price to the current closing price.

using System;

class Stock

{

 string stockName;

 string stockSymbol;

 double previousClosingPrice;

 double currentClosingPrice;

 public Stock(string name, string symbol, double previousPrice, double currentPrice)

 {

 stockName = name;

```

        stockSymbol = symbol;
        previousClosingPrice = previousPrice;
        currentClosingPrice = currentPrice;
    }

    public double GetChangePercentage()
    {
        return ((currentClosingPrice - previousClosingPrice) /
previousClosingPrice) * 100;
    }

    public void Display()
    {
        Console.WriteLine("\n----- Stock Details -----");
        Console.WriteLine("Stock Name      : " + stockName);
        Console.WriteLine("Stock Symbol      : " + stockSymbol);
        Console.WriteLine("Previous Close Price : " +
previousClosingPrice);

        Console.WriteLine("Current Close Price  : " +
currentClosingPrice);

        Console.WriteLine("Change Percentage    : " +
GetChangePercentage() + "%");
    }
}

```

```
class Program
{
    static void Main()
    {
        Console.Write("Enter Stock Name: ");
        string name = Console.ReadLine();

        Console.Write("Enter Stock Symbol: ");
        string symbol = Console.ReadLine();

        Console.Write("Enter Previous Closing Price: ");
        double previousPrice =
        Convert.ToDouble(Console.ReadLine());

        Console.Write("Enter Current Closing Price: ");
        double currentPrice =
        Convert.ToDouble(Console.ReadLine());

        Stock stock = new Stock(name, symbol, previousPrice,
        currentPrice);
```

```
        stock.Display();

        Console.ReadLine();
    }
}
```

OUTPUT:

Enter Stock Name: Reliance

Enter Stock Symbol: REL

Enter Previous Closing Price: 2500

Enter Current Closing Price: 2400

----- Stock Details -----

Stock Name : Reliance

Stock Symbol : REL

Previous Close Price : 2500

Current Close Price : 2400

Change Percentage : -4%

3) Create a new class called "RandomHelper" which contains the following:

- 1. A static method called RandInt() that accepts two integers and returns a random integer between**

them. The numbers should be inclusive (i.e., if you call `RandInt(1,10)` you should be able to generate both 1 and 10).

2. A static method called `RandDouble()` that accepts two integers and returns a random double between them. For this method you should be able to generate numbers such that $1 \leq x < 10$ for the method call `RandDouble(1,10)`.

3. Call your methods from another class without instantiating the class (i.e., call it just like `Math.Random()` since your methods are defined to be static).

```
using System;
```

```
class RandomHelper
```

```
{
```

```
    static Random r = new Random();
```

```
    public static int RandInt(int min, int max)
```

```
    {
```

```
        return r.Next(min, max + 1);
```

```
    }
```

```
    public static double RandDouble(int min, int max)
```

```
    {
```

```
        return min + r.NextDouble() * (max - min);
```

```
    }  
}  
  
class Program  
{  
    static void Main(string[] args)  
    {  
        Console.WriteLine("Random Integer : " +  
RandomHelper.RandInt(1, 10));  
  
        Console.WriteLine("Random Double : " +  
RandomHelper.RandDouble(1, 10));  
  
        Console.ReadLine();  
    }  
}
```

OUTPUT:

Random Integer : 3

Random Double : 2.59575930358645